

Capital Consistency under Regulatory Credit-Risk Uncertainty: Allocation Identities, Regulatory Kinks, and Joint Conservatism

Maarten Linus Herrmann
Erasmus School of Economics
Erasmus University Rotterdam
ORCID: [0009-0005-8721-6588](https://orcid.org/0009-0005-8721-6588)

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Abstract

We study two connected questions in the European credit-risk framework for internal ratings: when the allocation of a fixed economic history between exposure at default/credit conversion factor and loss given default leaves expected loss and risk-weighted assets unchanged, and how to define a minimum statistically valid joint margin of conservatism for PD, LGD, CCF, and allocation uncertainty. The elementary product identity is not new: under a performing-exposure IRB branch that is linear in LGD, a loss-consistent allocation with $Ag = L$ eliminates EAD–LGD choice. Current EU rules and EBA material already anticipate the corresponding post-default drawing compensation. The useful results are boundary results. We give exact moment and common-atom criteria, prove a sharp endpoint reduction after common LGD floors/clipping, identify an exact polytope projection for individual interval CCFs, and give rational counterexamples for binding floors and default-weighted calibration. For joint conservatism, we separate validity of a confidence set from deterministic worst-case optimization, show why component-wise bounds are sufficient only under simultaneous coverage and target monotonicity, and record that the corporate unexpected-loss kernel is not globally increasing in PD. A unified collapse-or-endpoint result shows when allocation uncertainty disappears, when it reduces to maximum EAD, and when coupling to grades, weights, or segments prevents separation. The package contains standard-library implementations, exact certificates, an independent IRB formula checker, and deterministic adversarial tests. The result is a technical note and research platform, not a claim that generic projection inference or EAD–LGD compensation is novel.

1 Motivation and claim boundary

For retail exposures, CRR3 allows future drawings to be reflected in conversion factors or LGD, with matching treatment in the LGD numerator and denominator when drawings enter conversion factors; non-retail post-default drawings must be reflected in IRB-CCF [6]. EBA/GL/2017/16 already explains the compensation principle for realised LGD and post-default drawings [4]. The pending/consultative CCF methodology has made allocation, multiple defaults, and post-default drawings operationally salient, but consultation text is not binding law [5]. The contribution cannot therefore be “equal products give equal capital.” Nor is maximizing a scalar over a joint confidence set new; projection inference, support functions, parameter-uncertainty capital, and distributionally robust optimization are extensive literatures [8, 3, 9, 2].

The defensible objective is narrower: formally identify the assumptions under which the regulatory calculation pipeline respects the product identity, construct small exact boundary counterexamples, and integrate allocation ambiguity into a precisely defined capital-valid confidence problem.

2 Fixed-history allocation model

Fix an economic history h containing commitments, drawn balances, signed discounted cash flows, recoveries and costs, default/cure/probation states, episode labels, and observation-unit labels. A rule δ maps h into records $i \in I_\delta$ with drawn exposure D_i , undrawn amount U_i , CCF q_i , EAD $A_i = D_i + q_i U_i > 0$, economic-loss numerator $L_i \geq 0$, LGD g_i , PD p_i , maturity M_i , segment s_i , and nonnegative weight w_i . A loss-consistent record satisfies

$$g_i = L_i/A_i. \quad (1)$$

This is a construction assumption. Separately fitted CCF and LGD models need not satisfy it observation by observation or after aggregation.

For the applicable performing-exposure IRB branch, write $H(p, M, s)$ for the coefficient excluding LGD, EAD, and the capital-to-RWA factor. Then

$$\text{EL}(\delta) = \sum_i w_i p_i A_i g_i, \quad \text{RWA}_0(\delta) = 12.5 \sum_i w_i A_i g_i H(p_i, M_i, s_i). \quad (2)$$

The implemented corporate test functional uses the current Basel/CRR3-style correlation and maturity expressions; legal applicability by exposure class remains separate [1, 7].

3 Allocation results

Theorem 1 (Exact moment criteria). *On the domain above, EL is invariant over an allocation class if and only if $\sum_i w_i p_i L_i$ is constant. Unadjusted performing-exposure RWA is invariant if and only if $\sum_i w_i L_i H(p_i, M_i, s_i)$ is constant.*

Proof. Substitute (1) into (2). Necessity and sufficiency are immediate because the equalities hold record by record. $\square$

The structural content appears after allocations are lifted to common economic atoms. Let $x(\delta)$ be the vector of weighted loss masses and let

$$V = \text{span}\{x(\delta) - x(\delta') : \delta, \delta' \in \mathcal{D}(h)\}.$$

A fixed coefficient vector c yields invariant $c^\top x(\delta)$ exactly when $c \in V^\perp$. If arbitrarily small loss transfers are permitted along a graph, c must be constant on each transfer-connected component. Total loss conservation alone is not enough when transfers cross PD, maturity, segment, or weight coefficients.

For lower LGD floor ℓ and upper clip u , $0 \leq \ell \leq u$, the operational loss product is

$$\psi_{\ell, u}(A, L) = A \min\{u, \max(\ell, L/A)\} = \min\{uA, \max(\ell A, L)\}. \quad (3)$$

It is nondecreasing in A and piecewise linear, but generally neither convex nor concave when both bounds matter.

Proposition 1 (Sharp endpoint reduction). *Let $A(q) = D + U \max(q, q_f) > 0$, with $U \geq 0$ and $q \in [q_-, q_+]$. For fixed L, ℓ, u and a fixed nonnegative IRB coefficient, the minimum and maximum adjusted capital occur at q_- and q_+ respectively.*

Proof. Both $A(q)$ and (3) are nondecreasing. The endpoints are feasible. $\square$

If finite total facility amount $B = D + U \geq 0$ is fixed and individual finite $q \in [q_-, q_+]$, introduce converted undrawn exposure $X = qU$. After eliminating q , the exact raw feasible projection is

$$D + U = B, \quad 0 \leq U \leq B, \quad q_- U \leq X \leq q_+ U, \quad A = D + X, \quad (4)$$

a polytope with $A \in [B \min(1, q_-), B \max(1, q_+)]$. A realised-LGD domain requiring $A > 0$ must intersect this raw set with that strict inequality (or impose $A \geq \varepsilon > 0$ to retain a closed polytope). The full graph retaining $X = qU$ is bilinear and generally nonconvex. For a CCF shared across records, the exact graph remains nonconvex; under a CCF-independent base polytope, however, its convex hull and every unrestricted linear support problem reduce exactly to the two CCF endpoint graphs.

There is also a sharp allocation-only support bound when conserved nonnegative loss is the sole restriction. If $x_i \geq 0$, $\sum_i x_i = L$, and the coefficients c_i are fixed, then

$$L \min_i c_i \leq c^\top x \leq L \max_i c_i. \quad (5)$$

Both bounds are attained by concentrating all loss on an extreme-coefficient atom. More generally, for a nonempty coefficient set C with finite coordinate supports,

$$\sup_{c \in C, x \geq 0, \mathbf{1}^\top x = L} c^\top x = L \max_i h_C(e_i). \quad (6)$$

The case $L = 0$ is handled separately and has value zero. These classical linear-programming bounds quantify the largest possible failure of common-atom invariance when only total loss is conserved.

Example 1 (Binding LGD floor). *Fix $D = U = 100$ and $L = 10$. The allocations $q = 0$ and $q = 1$ give $(A, g) = (100, 1/10)$ and $(200, 1/20)$, preserving $Ag = 10$. An LGD floor $1/10$ gives operational products 10 and 20. This is an exact rational witness.*

This and the weighting example below are abstract mathematical witnesses motivated by regulatory-type operators, not complete legally co-feasible exposure-class examples under the full current EU operator stack.

Example 2 (Default weighting). *Two facilities each lose 10. EAD vectors $(100, 100)$ and $(50, 150)$ both preserve each facility's loss and total EAD. Default-count-weighted mean LGD followed by application to total EAD gives 20 for the first allocation and $80/3$ for the second. Exposure weighting gives 20 for both. Thus calibration and aggregation order are part of the invariance problem.*

An exact named decomposition of a capital difference is available by Boolean-lattice Möbius inversion only after valid baseline/target hybrid calculations are defined. The construction depends on the baseline and category definitions and selects no baseline-free split without additional axioms; no universal impossibility result for every other attribution rule is claimed.

4 Joint margin of conservatism

Let η collect PD, LGD, CCF, allocation, weights, segments, deficiencies, and floor status. Let C_α be a joint confidence region or deterministic uncertainty set with an explicitly stated interpretation. For target risks R_j , an operational choice $\tilde{\eta}$ is valid relative to C_α when

$$R_j(\tilde{\eta}) \geq \sup_{\eta \in C_\alpha} R_j(\eta) \quad \text{for every required } j. \quad (7)$$

The construction/coverage of C_α and optimization in (7) are separate validity layers. Projection of a full joint region can be conservative relative to target-calibrated inference [8].

For one scalar target and a fixed valid set, put $b = R(\hat{\eta})$, $M = \sup_{C_\alpha} R$, $V = \{R(t) : t \in \mathcal{T}\}$, and $v_* = \inf(V \cap [M, \infty))$ when that set is nonempty. The infimum nonnegative risk-scale uplift is $[v_* - b]_+$. If V contains $\max\{b, M\}$, this is the attained value

$$\max\{0, \sup_{\eta \in C_\alpha} R(\eta) - R(\hat{\eta})\}. \quad (8)$$

If the threshold is only approachable, the value is an infimum rather than an attained minimum. Equation (8) is conditional on the set and does not identify a unique parameter vector or a globally shortest statistical procedure.

Separate parameter bounds are sufficient when the joint component-wise upper event has coverage at least $1 - \alpha$ and every target is coordinatewise nondecreasing on the relevant domain. Marginal 95% bounds have joint coverage $0.95^2 = 0.9025$ under independence in two dimensions. With arbitrary dependence, Bonferroni is valid and the union bound is sharp given only marginal failure bounds.

For affine $R(\theta) = a^\top \theta + b$ and ellipsoid $C = \{\mu + \Sigma^{1/2}z : \|z\|_2 \leq r\}$,

$$\sup_{\theta \in C} R(\theta) = a^\top \mu + b + r\sqrt{a^\top \Sigma a}. \quad (9)$$

This classical support formula remains valid for singular $\Sigma \succeq 0$; it is not exact for a nonlinear or kinked target merely by substituting a gradient.

Category-wise affine margins add exactly only under a compatible joint geometry. If each category set $\mathcal{U}_k$ and a nonempty coupling set $\Gamma \subseteq \prod_k \mathcal{U}_k$ are compact, then

$$\sup_{e \in \Gamma} R\left(\hat{\theta} + \sum_k e_k\right) \leq R(\hat{\theta}) + \sum_k h_{\mathcal{U}_k}(a). \quad (10)$$

Equality holds exactly when Γ contains a tuple simultaneously attaining every category support. For nonlinear targets, the analogous one-at-a-time increments add exactly only when the total higher-order Möbius interaction is zero; monotonicity alone does not sign the error.

Most importantly, the corporate unexpected-loss kernel is not globally increasing in PD. As $p \uparrow 1$, its conditional-default term and p both tend to one, so their difference and the capital coefficient tend to zero, while the coefficient is positive at ordinary interior PD. In the implementation with LGD 0.45 and maturity 2.5, $k(0.30) \approx 0.19905$ but $k(0.90) \approx 0.04327$. Upper PD can therefore lower RWA even though it raises expected loss. A direct capital bound must optimize the actual target.

For one performing record this joint EL/RWA action has an exact one-dimensional reduction. Let effective loss $Y \in [\underline{Y}, \bar{Y}]$, PD $p \in P$, and positive continuous unexpected-loss coefficient $\kappa(p)$. Thresholds $pY \geq M_E$ and $12.5\kappa(p)Y \geq M_K$ reduce to global search over

$$Y^*(p) = \max \left\{ \underline{Y}, \frac{M_E}{p}, \frac{M_K}{12.5\kappa(p)} \right\}, \quad (11)$$

restricted to $Y^*(p) \leq \bar{Y}$. This preserves PD nonmonotonicity. Given only finite marginal failure probabilities, the sharp lower simultaneous coverage is the Bonferroni bound; with finite parameter marginals and unknown coupling, worst expected capital is a multi-marginal transport linear program. These are information limits, not a new estimator.

5 Unified collapse-or-endpoint result

Suppose nonempty statistical and allocation domains are given, statistical uncertainty determines a nonnegative coefficient $c(\theta)$ with finite supremum, and allocation determines only $A(q)$, with fixed loss L . Then

$$R(\theta, q) = c(\theta)\psi_{\ell, u}(A(q), L). \quad (12)$$

If no floor or clip binds, allocation uncertainty collapses exactly because $\psi = L$. With common floor/clip and interval CCF uncertainty,

$$\sup_{\theta \in C, q \in [q_-, q_+]} R(\theta, q) = \left(\sup_{\theta \in C} c(\theta) \right) \psi_{\ell, u}(A(q_+), L). \quad (13)$$

Thus the joint supremum factorizes and its allocation part reduces to the upper CCF endpoint in this separable nonnegative class. Iterated suprema commute for every functional on a fixed Cartesian product. Coupling allocation to grade, PD, maturity, segment, weights, or the confidence set can instead break the product factorization and endpoint reduction. Generic sup/inf minimax interchange additionally requires compactness and convexity/concavity; discrete segmentation and regulatory kinks do not supply these assumptions.

6 Computational evidence and audit

The standard-library package contains versioned history records, exact rational allocation objects, the current corporate IRB test formula, a fully independent normal-CDF/inverse implementation, ellipsoid and box support functions, endpoint optimization, certificate producers, and independent semantic verifiers. The test suite covers exact enumeration, malformed and non-finite inputs, repeated episodes, extreme-scale singular covariance, formula cross-checks, floors, weighting, and PD nonmonotonicity. Three certificates verify the floor witness, weighting witness, and affine ellipsoid support; the first two are rational, while the ellipsoid witness is a closed-form floating certificate with a stored factor, feasible primal witness, and matching Cauchy upper bound.

A deterministic adversarial atlas with seed 20260806 tests 5,000 random fixed-loss reallocation pairs and a 20,001-point PD grid. These are finite numerical regression results, not proofs or certified global optimization. Proof status rests on the algebra; exact-rational status applies only to the rational witnesses.

The canonical audit command is

```
bash scripts/audit_repository.sh.
```

For release v0.1.0 it passes 36 unit tests, independently verifies all three certificates, reproduces every deterministic experiment and manifest, and performs boundary, symlink, cleanup, and size checks. The public repository distributes the reviewed PDF but intentionally omits the LaTeX working source; the software and certificate audit itself is local and network-free. Release materials are available at <https://github.com/Maar10Herr/Capital-Consistency-Public>.

7 Assessment and open work

The allocation workstream survives as a narrowly scoped technical contribution. The simple identity and regulatory compensation principle are antecedents; the exact boundary atlas, common-atom criterion, polytope reduction, and small exact counterexamples are useful. The joint-MoC workstream survives as a research program, but current scalar/ellipsoid results are classical. A genuinely stronger statistical contribution would require target-calibrated finite-sample or uniform-asymptotic coverage for the exact kinked EU-IRB functional with dependent estimators and stable treatment of singular covariance.

The unified collapse-or-endpoint result is the strongest current synthesis. It precisely separates regimes where allocation ambiguity is irrelevant, endpoint-solvable, or inseparable from statistical and structural uncertainty. Independent mathematical, optimization, and computational audits informed the final scope. Before regulatory or production application, the legal operator order and applicable exposure classes still require external specialist review.

Current assessment.

Workstream	Status	Open limitation
Allocation invariance	survives narrowly	Novelty is in boundary characterization, not the product identity.
Joint margin of conservatism	open program	No estimator-specific joint coverage theorem for the kinked target.
Unified framework	strongest synthesis	Coupled full-portfolio optimization is not globally certified.
Regulatory mapping	research-grade map	External legal review is still required.

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